Power BI Project

Telco Customer Churn Analysis Dashboard Using Power BI

Explanation

This project report describes the supplied Telco Customer Churn dashboard. It follows the section order of the provided reference report while using only details supported by the supplied PBIX.

Overview

The dashboard is a single-page Power BI report built around the Telco_Churn model entity. It brings customer and churn KPIs together with charts that compare churn rate across contract, tenure, internet service, and payment method.

The saved report includes two interactive slicers, for Internet Service and Contract. Selecting these categories lets a viewer explore how the displayed measures change in the report.

Objectives

• Summarize customer and churn measures in one view.

• Compare churn rate across the categories shown in the dashboard.

• Explore report results with Internet Service and Contract slicers.

• Present the report in a clear, documented Power BI project.

No student name or roll number is included because those details were not verified for this dashboard.

Tools & Technologies

Microsoft Power BI Desktop

DAX (Data Analysis Expressions) measure references saved in the PBIX

Power BI report visuals and slicers

Embedded Power BI data model

Dashboard Features

The dashboard provides a single-page view of customer and churn metrics. Its visual analysis includes:

Total Customers and Churned Customers

Churn Rate and Average Monthly Charge

Churn rate by Contract

Churn rate by Tenure Group

Churn rate by Internet Service

Churn rate by Payment Method

The saved report contains two selectable slicers: Internet Service and Contract.

Dataset

The PBIX contains an embedded model entity named **Telco_Churn**. The original source dataset or provider, full model schema, row count, and refresh date could not be verified from the readable report definition.

There is no separate CSV or source-data file in the supplied project materials.

Main Attributes

The fields directly bound to report visuals are:

Contract

Internet Service

Tenure Group

Payment Method

The KPI cards also reference four model measures: Total Customers, Churned Customers, Churn Rate, and Avg Monthly Charge.

Dashboard Architecture

The dashboard is one page named Page 1 on a 1920 x 1080 canvas. A title text box sits in the upper-left area. Four KPI cards occupy the upper area, with Internet Service and Contract slicers at the upper-right. The lower report area contains a clustered bar chart for Contract, a line chart for Tenure Group, a clustered column chart for Internet Service, and an area chart for Payment Method.

Project Workflow

Embedded Telco_Churn model -> Report visual bindings -> KPI cards and charts -> Filter with the two slicers

This sequence summarizes the saved report structure. The data cleaning, transformation, relationship design, and refresh workflow are not documented in the PBIX definition that could be inspected.

DAX

The report definition confirms references to the following measures:

Total Customers

Churned Customers

Churn Rate

Avg Monthly Charge

Their DAX formulas, calculation logic, and business definitions could not be extracted from the compressed model. The report names the measures but does not infer formulas.

Dashboard Insights

A text box saved on the dashboard states that month-to-month customers churn at 43% versus 27% overall, and customers in their first year reach 51%. It also suggests an early contract-upgrade offer.

These numbers and the recommendation are included as existing dashboard annotation text. They were not independently recalculated or validated against the embedded model.

The visuals are intended to help viewers compare churn rate across Contract, Tenure Group, Internet Service, and Payment Method, then explore the KPI cards with the two slicers.

Dashboard

Schematic layout reconstructed from the PBIX visual definitions. It is not a screenshot of the live Power BI page, and it does not show report values.

Project Files

Telco-Customer-Churn-Dashboard/

README.md

Telco-Customer-Churn-Dashboard.pbix

Telco-Customer-Churn-Project-Report.pdf

PBIX File

The PBIX contains the report layout, visual bindings, and embedded DataModel. The original data source is not included as a separate file. Measure names are visible in the report definition, while their DAX formulas could not be verified.

How to Use

Open Telco-Customer-Churn-Dashboard.pbix in Microsoft Power BI Desktop.

How to Use

1. Download or open the project from the GitHub folder linked on the final page.
2. Install or open Microsoft Power BI Desktop.
3. Open Telco-Customer-Churn-Dashboard.pbix.
4. Select Internet Service and/or Contract slicer values to filter the page.
5. Read the KPI cards and compare the four churn-rate charts.
6. Clear the slicer selections to return to the unfiltered view.

What the Dashboard Helps to Show

The report is designed to help viewers ask:

How do the customer and churn KPIs change with the slicer selection?

How does churn rate compare across contract types and tenure groups?

How does churn rate vary across internet service and payment method?

Which categories should be examined further in the live report?

Project Purpose

This project documents the supplied Power BI report and demonstrates an interactive approach to exploring telco churn measures by customer categories.

Conclusion

The Telco Customer Churn dashboard brings four headline measures, four category comparisons, and two slicers together on one page. Its report structure is identifiable from the PBIX. The source dataset, DAX formulas, and numeric results are not independently verifiable from the available report definition, so this document separates the dashboard content from verified findings.

GitHub Link:

https://github.com/neeraj-gitthub/neeraj-gitthub/tree/main/projects/telco-customer-churn-dashboard